

# Empirical Proof Record: VALIDATED

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## Empirical Proof Record — VALIDATED

**Proof ID:** ab9f13103a03b2257566ffb4a090d1b0...

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### 1. Claim

Across 30 synthetic two-sample pairs of size 50 per group with varying effect sizes and variance ratios (seed=20260518), “`scipy.stats.ttest_ind(equal_var=False)`“, “`statsmodels.stats.weightstats.ttest_ind(usevar='unequal')`“, and “`pingouin.ttest(correction=True)`“ produce Welch t-statistics AND two-sided p-values that agree to  $\leq 1e-09$  across all pairwise comparisons. (RFC 0002 Phase E1: cross-framework validation; three-way variant with genuinely-independent statsmodels.)

- **Operationalisation:** Generate  $N$   $(x, y)$  pairs where  $x \sim \text{Normal}(0, \sigma_x^2)$  and  $y \sim \text{Normal}(\delta, \sigma_y^2)$  for a sweep of  $\delta$  across  $[-1.0, 1.0]$  and variance-ratio  $\sigma_y/\sigma_x$  across  $[0.5, 2.0]$ . Compute Welch’s  $t$  and two-sided  $p$  under each backend. Compute  $\text{max\_abs\_diff} = \max \text{ over } (\text{pair}, \text{statistic} \in \{t, p\}, \text{backend-pair}) \text{ of } |\text{stat\_a} - \text{stat\_b}|$ . VALIDATED iff  $\text{max\_abs\_diff} \leq \text{tolerance}$ .
- **Threshold:**  $\text{max\_absolute\_welch\_difference} \leq 1e-09$   $t_{\text{or}} p$
- **H0:** At least one backend pair disagrees on the Welch  $t$ -statistic OR the two-sided  $p$ -value by more than the floating-point tolerance.
- **H1:** All three backends compute identical Welch  $t$  AND  $p$  to floating-point tolerance across every pair.

### 2. Verdict

**VALIDATED** — observed  $1.7763568394002505e-15$

30 pairs  $\times$  3 backends  $\times$  2 statistics; max pairwise  $|\Delta| = 1.776e-15$  on  $(t, \text{scipy\_vs\_statsmodels})$  ( $\leq 1e-09$  tol)

### 3. Evidence

| Pillar                                   | Statistic                     |                   |
|------------------------------------------|-------------------------------|-------------------|
| welch_t_scipy_vs_statsmodels_vs_pingouin | max_absolute_welch_difference | 1.776356839400250 |

### 4. Pre-registration

- Registered at: 2026-05-24T03:31:24.540632+00:00
- Config hash: 650f785bb6c4c650aa559236e9e29db3...
- Data hash: 650f785bb6c4c650aa559236e9e29db3...

### 5. Signature

ab68516bc329773a9921e60c061dde7b99b45a6e4651d440aad33c501270def2